AP CALCULUS AB - UNIT 5

Guided Notes 5.2–5.5 II: Derivative Graph Analysis - Part 2: Challenge Questions

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we determine intervals of concavity and locate points of inflection?

- Determine intervals of concavity and locate points of inflection.
- Use the Second Derivative Test when its hypotheses are satisfied.
- Relate the graphs and signs of f, f', and f''.
- Synthesize first- and second-derivative information in complete curve analysis.

Key Knowledge, Vocabulary, and Notation

Concave up/down — $f'' > 0$ means slopes f' increase; $f'' < 0$ means slopes decrease.

Inflection point — A point on the graph where concavity changes; $f'' = 0$ alone is only a candidate.

Second Derivative Test — If $f'(c) = 0$, then $f''(c) > 0$ gives a local min and $f''(c) < 0$ gives a local max; $f''(c) = 0$ is inconclusive.

Graph of f' — Where f' increases, f is concave up; where f' decreases, f is concave down.

Graph of f'' — Its sign directly determines concavity of f and increase/decrease of f'.

Complete analysis — Combine domain, intercepts, asymptotes, critical points, monotonicity, concavity, and end behavior.

Concept Development and Strategy

1. Solve $f'' = 0$ and include points where f'' is undefined, then test concavity on the resulting intervals.
2. Verify an actual sign change in concavity before naming an inflection point.
3. When reading f', distinguish its sign (monotonicity of f) from its slope (concavity of f).
4. The Second Derivative Test applies only at a critical point with a suitable second derivative sign.
5. A complete AP explanation links the derivative evidence to a precise conclusion about the original function.

Shared Representation / Data

Interval	$(-\infty, -2)$	-2	$(-2, 1)$	1	$(1, \infty)$
$f'(x)$	$+$	0	$-$	0	$+$
$f''(x)$	$-$		$-$ then $+$		$+$

Worked Example: Concavity

For $f(x) = x^4 - 4x^3$, determine concavity and inflection points.

Answer and Reasoning

$f'' = 12x(x-2)$. Concave up on $(-\infty, 0) \cup (2, \infty)$ and down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$, with points $(0, 0)$ and $(2, -16)$.

Worked Example: Second Derivative Test

Classify critical points of $f(x) = x^3 - 3x^2 + 2$ using f''.

Answer and Reasoning

$f' = 3x(x-2)$ gives $x = 0, 2$; $f'' = 6x - 6$. $f''(0) = -6$, local max; $f''(2) = 6$, local min.

Worked Example: From the graph of f'

If f' is positive and decreasing on $(1, 4)$, describe f.

Answer and Reasoning

f is increasing because $f' > 0$ and concave down because f' is decreasing.

Worked Example: Full sign synthesis

Using the shared table, describe the behavior at $x = -2$ and $x = 1$.

Answer and Reasoning

The f' sign changes $+ \to -$ at -2, giving a local maximum. It changes $- \to +$ at 1, giving a local minimum. Concavity information must be read from the f'' row on neighboring intervals.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. If $f'' > 0$ on an interval, describe concavity.

2. If f' is decreasing, what is the sign of f''?

3. For $f(x) = x^3 - 6x$, find the inflection point.

4. For $f''(x) = (x - 2)(x + 1)$, determine where f is concave down.

5. For $f(x) = x^4 - 8x^2$, find local extrema and inflection points.

6. For $f'(x) = x^2e^{-x}$, determine monotonicity and concavity of f.

7. Construct a function with $f''(0) = 0$ but no inflection at 0.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. An inflection point requires

(A) $f'' = 0$ only
(B) a local extremum
(C) a change in concavity
(D) $f' = 0$

2. If $f' > 0$ and $f'' < 0$, then f is

(A) increasing and concave down
(B) decreasing and concave down
(C) increasing and concave up
(D) decreasing and concave up

3. For $f(x) = x^4 - 4x^2$, an inflection input is closest to

(A) 0
(B) 0.58
(C) 0.82
(D) 1.41

Common Errors and AP Precision

- A zero derivative or zero second derivative is a candidate condition, not by itself a complete extremum or inflection justification.
- State theorem hypotheses, include endpoints in absolute-extrema work, and verify every optimization candidate on the feasible domain.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

Construct a function with an inflection at 0 where $f''(0)$ does not exist.